2. a. Explain why consumers buy less of a good as its price rises.
   b. Discuss four factors that can shift the demand curve.

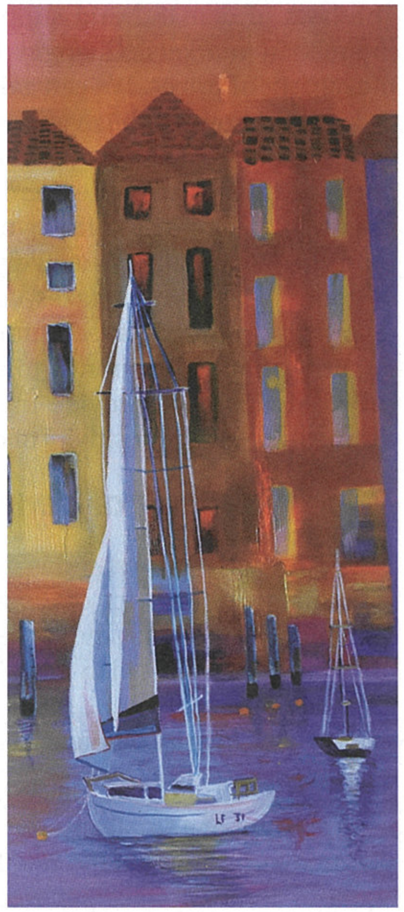
3. a. Explain why there is a positive relationship between price and quantity supplied.
   b. Discuss four factors that can shift the supply curve.

4. a. Explain the meaning of the term 'market equilibrium'.
   b. Explain why surpluses and shortages may exist, and why they are eliminated in competitive markets.
   c. Distinguish between an increase in demand, and an increase in quantity demanded.

5. Assume that a disease destroys much of the coffee crop in Brazil. Use the demand-supply model to explain the impact on the coffee market. How will the tea market be affected? What will happen to tea prices? Explain your answer using demand-supply analysis.

6. Explain each of the following using the model of demand and supply:
   a. The demand for computers increases over time and yet computer prices continually fall.
   b. The price of fruit and vegetables increases after a flood.
   c. Bicycle sales soar with rising petrol prices.
   d. Consumers are willing to pay more for organic products.

3 Elasticity



Learning Objectives

In this chapter you will learn about:

- ***the concept, and measurement, of price elasticity of demand***
- ***the determinants of price elasticity of demand***
- ***the link between price elasticity of demand and total revenue***
- ***the concept, and measurement, of income elasticity of demand***
- ***the concept of cross elasticity of demand***
- ***the concept of price elasticity of supply***
- ***the determinants of price elasticity of supply***
- ***the significance of price and income elasticity for consumers, business and government***

The laws of demand and supply help to explain how consumers and producers interact in markets. Changes in demand and supply also help to explain the movement of prices. We can gain further understanding about demand and supply by knowing how responsive consumers and producers are to a given change in price. For example, if the price of pizza were to increase by 10 per cent, would consumers decrease their quantity demanded by more than 10 per cent, less than 10 percent, or exactly 10 per cent? In other words, it is useful to know whether the law of demand (and the law of supply) is likely to be relatively strong or weak for different types of goods. This chapter takes us behind the demand curve and the supply curve to provide more detailed information about how much quantity demanded or quantity supplied responds to a given change in price. Economists use a concept called **elasticity** to provide this information. Price elasticity is a measure of the responsiveness or sensitivity of quantity to a change in price.

Price elasticity of demand

Price elasticity of demand is defined as the responsiveness of quantity demanded to a change in the price of the good or service. To help illustrate price elasticity of demand, suppose that the present price of a coffee (e.g. cappuccino) is $4, and one million coffees are sold in Perth every week. What will happen to the quantity sold if the price rises by 10 per cent or $0.40? We know from the law of demand that when price rises, quantity demanded falls. But by how much will sales fall?

As a Year 11 economist, would you predict that coffee consumption would fall by a large amount or a small amount after the 10 per cent price rise? Suppose that market research indicated that the $0.40 price rise would lead to a fall in quantity demanded equal to 30,000 coffees per week. We need to know how large the fall in quantity was relative to the price increase. To do this we compare the percentage change in price with the percentage change in quantity. The quantity change is 30,000 which represents a 3 per cent (30,000 /1 million) change. So, a ten per cent price rise caused quantity demanded to fall by just 3 per cent! From this evidence we could conclude that the demand for coffee is not very responsive to changes in price. We would say that the demand for coffee is **inelastic** or not responsive.

We can calculate price elasticity of demand (Ed) using a number of methods. If we use the percentage change method, the formula is:

$$Ed = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

In our example, $Ed = 3\%/10\% = 0.3$.

As long as the percentage change in quantity is less than the percentage change in price, the answer will be less than one and this indicates that demand is inelastic. In this example, it was relatively easy to calculate the percentage change, but in other examples, the data may be more complex. In such cases, it is easier to use an alternate formula:

$$Ed = \frac{\Delta Q}{Q} / \frac{\Delta P}{P}$$

where Q = quantity demanded; P = price and the delta symbol Δ refers to 'change'.

The equation can be transformed to

$$Ed = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P}$$

Substituting the figures from the coffee example above,

$$Ed = \frac{30{,}000}{1{,}000{,}000} \times \frac{4}{0.4}$$

$$= 0.3$$

The answer is known as the **elasticity coefficient**. In this instance it means that a 1 per cent change in the price of coffee will lead to a 0.3 per cent change in quantity demanded. When the value is less than 1, it means that the law of demand is relatively weak - quantity demanded is not very responsive to a price change. Why not? It is usually determined by the substitution effect. Coffee is addictive and has few close substitutes. If the price of coffee rises then consumers cannot easily substitute away from the product. Demand will be relatively inelastic. Other goods with inelastic demand would be alcohol (beer wine, spirits), petrol and basic food products such as bread and milk. Notice that goods that are necessities will have few close substitutes and will therefore be inelastic. This means that quantity demanded responds weakly to a change in price.

The price elasticity coefficients are sometimes shown with a minus sign in front of them because quantity and price are negatively related (the demand curve has a negative slope). By convention, however, we ignore the minus sign because we are only interested in the absolute number.

What about products that have relatively close substitutes. We would expect that quantity demanded would be more sensitive to a change in price. Suppose that the price of pizza were to rise relative to other fast food goods. Assume that the price of pizza were to increase from $10 to $11 and weekly pizza consumption decreased from 500,000 to 400,000. What would be the price elasticity of demand for pizza? Applying our formula:

$$Ed = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P}$$

$$Ed = \frac{100{,}000}{500{,}000} \times \frac{10}{1}$$

$$= 2.0$$

What does the answer mean? It means that a 1 per cent change in the price of pizza results in a 2 per cent change in the quantity demanded; or a 10 per cent change in price leads to a 20 per cent change in quantity demanded. When the value of Ed is greater than 1, it means that the law of demand is relatively strong - quantity demanded is responsive to a price change. In this case demand is said to be relatively **elastic**. The larger the elasticity coefficient, the more responsive demand is to price which indicates that there is likely to be many close substitutes.

Figure 3.1 compares a relatively elastic demand curve (D1) with an inelastic demand curve (D2). On both demand curves, price has fallen by 20 per cent - from \$10 to \$8. On demand curve D1, quantity increases from 100 to 150 - an increase of 50 per cent. But on demand curve D2, quantity increases from 100 to 110 - an increase of just 10 per cent. So D1 reflects an elastic demand with Ed = 2.5, while D2 reflects an inelastic demand with Ed = 0.5.

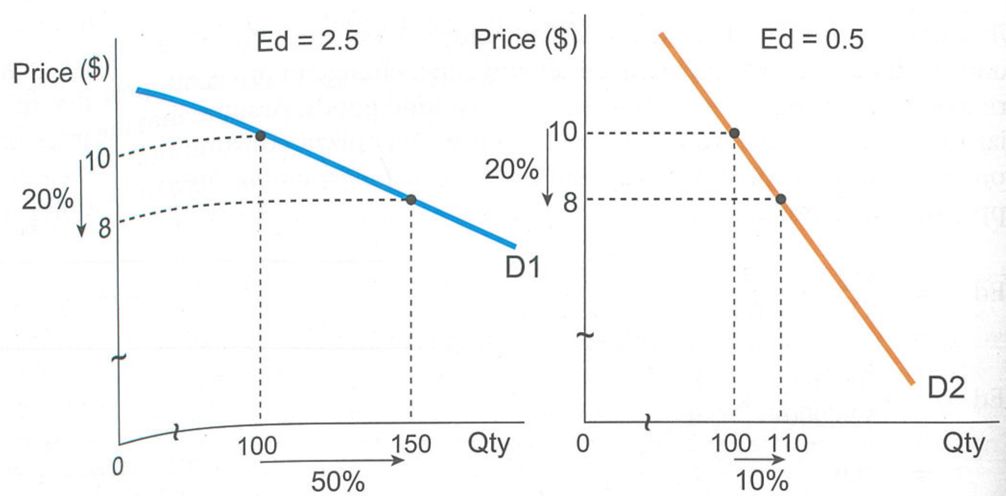
The formula we have been using is called the **point method** because we are calculating elasticity from a specific point or price on the demand curve. In the case of the pizza example, we calculated elasticity for a price increase from \$10 to \$11. But what if we calculated the elasticity for a price decrease from \$11 to \$10 (quantity rising from 400,000 to 500,000)? Would we get the same answer? The answer is no! The answer changes because the value for P and Q in our formula will change:

$$\mathbf{Ed} = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P}$$

$$\text{Ed} = \frac{100{,}000}{400{,}000} \times \frac{11}{1}$$

$$= 2.75$$

Figure 3.1 Elastic and inelastic demand



Using the simple point method gives two different answers depending on whether we are increasing or decreasing price. To avoid this confusion we use an averaging technique, called the **midpoint method**. In our formula we use the average or midpoint price and the average or midpoint quantity. This would mean that in our pizza example, the value for price would be \$10.50 (P_{ave}) and the value for quantity would be 450,000 (Q_{ave}). The two change variables (ΔP and ΔQ) stay the same.

Using the new midpoint formula:

$$\mathbf{Ed} = \frac{\Delta Q}{Q_{ave}} \times \frac{P_{ave}}{\Delta P}$$

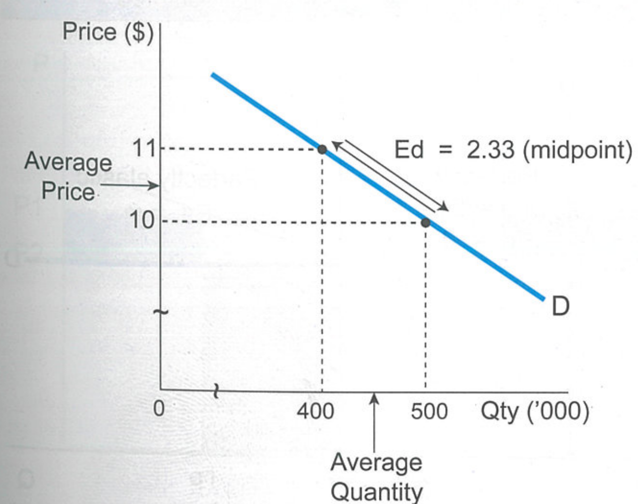
$$\text{Ed} = \frac{100{,}000}{450{,}000} \times \frac{10.5}{1}$$

$$= 2.33$$

The advantage of using the midpoint formula is that the answer does not depend on which price we choose as the starting price - instead we are calculating the elasticity along the same segment of the demand curve. The midpoint method for calculating price elasticity is shown below in figure 3.2. This is usually the preferred method used by economists.

Along any normal downward sloping demand curve, price elasticity decreases as we move down the demand curve. In other words price elasticity falls as price falls. This makes sense if you remember the formula: $Ed = \Delta Q / \Delta P \times P/Q$. As price falls, quantity rises so that the P/Q value will decrease. Generally consumers are not that price sensitive to relatively inexpensive goods and are more price sensitive to expensive luxury goods. For example, if a good that costs \$1 increases by 10 per cent to \$1.10 it will not have much impact on the spending decisions of buyers. But if a

Figure 3.2 The midpoint method



Price elasticity of demand measures the responsive of quantity demanded to a change in price. To measure the elasticity between \$10 and \$11 we use the midpoint formula - this means using the average price and the average quantity in the elasticity formula.

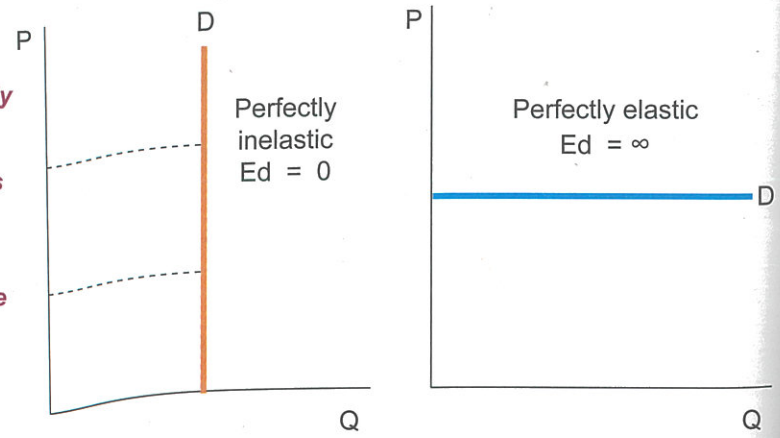
good that costs $5,000 increases in price by 10 per cent to $5,500 it is likely to impact on buyers significantly.

In theory, the value of price elasticity of demand can range between zero and infinity. If price elasticity equals zero, then a change in price will have no effect on quantity demanded - demand is said to be **perfectly inelastic** (Ed = 0). The demand for heroin by a drug addict or insulin for a diabetic would be examples of perfectly inelastic demand. If price increases, there is no change in quantity demanded - the law of demand in this case stops working. A perfectly inelastic demand curve is vertical. A horizontal demand curve is said to be **perfectly elastic** and price elasticity would equal infinity (Ed = ∞). An example of this would be a good with a perfect substitute - if the price of one good increases by one cent, consumers would stop buying the good and switch to the substitute. Figure 3.3 summarises the different types (values for price elasticity of demand. The slope of the demand curve is often used to convey whether demand is likely to be elastic or inelastic. Generally, the steeper the demand curve, the lower the elasticity (more inelastic) and the flatter the demand curve, the higher the elasticity (more elastic).

Figure 3.3 Price elasticity of demand

Type	Effect of a 1% increase in price	Examples
Perfectly inelastic Ed = 0	Quantity demanded does not change - the D curve is vertical	Goods with no substitutes
Inelastic Ed < 1	Quantity demanded decreases by less than 1% - relatively steep D curve	Goods with few close substitutes
Unitary elastic Ed = 1	Quantity demanded decreases by exactly 1% - equally proportional D curve	Goods with some close substitutes
Elastic Ed > 1	Quantity demanded decreases by more than 1% - relatively flat D curve	Goods with many close substitutes
Perfectly elastic Ed = ∞	Quantity demanded decreases to zero - horizontal D curve	Goods with perfect substitutes

When demand is perfectly inelastic, a change in price has no effect on Qd - the demand curve is vertical. When demand is perfectly elastic, a change in price causes an infinitely large change in Qd - the D curve is horizontal.



Price elasticity and total revenue/total expenditure

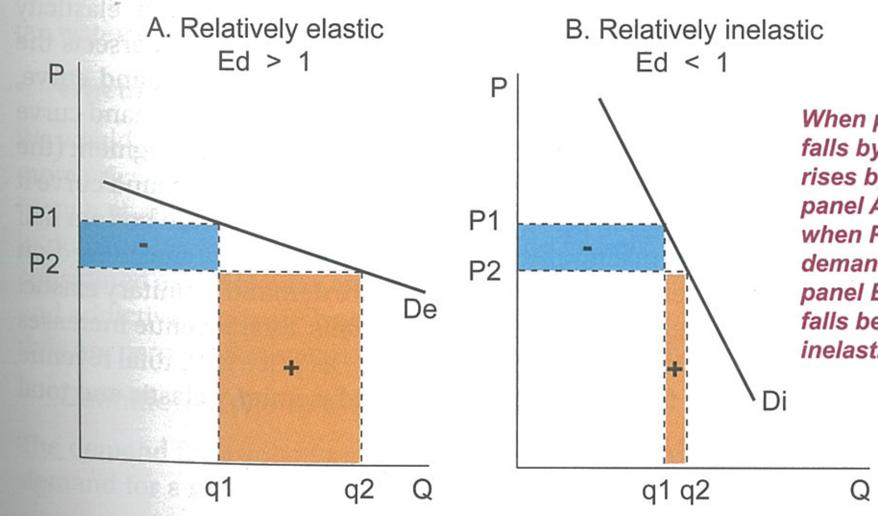
Price elasticity of demand is an important concept because of its link with the concepts of **total revenue** of business firms and **total expenditure** of households. Total revenue or total expenditure is simply price multiplied by quantity:

$$TR\ (TE) = P \times Q$$

Total revenue or expenditure can be measured using the demand curve. Figure 3.4 shows two demand curves - one that is relatively elastic (flat) and one that is relatively inelastic (steep). Total revenue (expenditure) is shown by the rectangle formed by price and quantity underneath the demand curve. If price falls from P1 to P2 then quantity demanded will increase from Q1 to Q2. But notice that the effect on total revenue is different. In panel (a) where demand is relatively elastic, total revenue has increased because the change in Qd is much greater than the change in price. In panel (b) where demand is relatively inelastic, total revenue has decreased because the change in Qd is much smaller than the change in price.

What would happen if we increased price? If demand is elastic, total revenue will fall, because the percentage decrease in Qd will be greater than the percentage increase in price. If demand is inelastic, then total revenue will increase - the fall in Qd will be smaller than the rise in price. So the important conclusion is that if we know the price elasticity of a good, then we can accurately predict what will happen to a firm's total revenue when it changes the price of its product. Why do firms have discount sales? Presumably to increase their revenue which means they think that the demand for their product must be elastic. What happens to total revenue (expenditure) if demand is unitary elastic and price changes? The answer is nothing, total revenue would not change. This is because if price increased by 10 percent, quantity demanded would decrease by exactly 10 per cent (Ed = 1).

Figure 3.4 Price elasticity and total revenue



When price falls, revenue falls by the (-) area and rises by the (+) area. In panel A, TR increases when P falls because demand is elastic. In panel B, TR falls when P falls because demand is inelastic.

Let's summarise the relationship between price elasticity of demand and total revenue:

- When D is elastic, Price and TR revenue move in opposite directions e.g. a rise in P decreases TR, while a fall in P increases TR
- When D is inelastic, Price and TR revenue move in the same direction e.g. a rise in P increases TR, while a fall in P decreases TR
- When D is unitary elastic, a change in P does not change TR.

The concept of elasticity plays an important role in a firm's pricing policy. Why do some firms (e.g. cinemas) charge children and students a lower price than adults? Why do hairdressers charge males a lower price than females for a haircut? The answer is that different consumer groups have a different price elasticity of demand and firms can use this information to charge different prices and increase their total revenue. This common pricing practice is known as **price discrimination** and is used by many firms throughout the economy. Firms can boost their revenue by segmenting their customers into separate groups according to their elasticity. Why do females pay more for hairstyling services? Because their demand for these services is very inelastic. By charging females a higher price than males, hair salons can increase their revenue. At the same time, the hair salons can also boost their revenue by charging males a much lower price because their demand is relatively elastic. Many firms charge a higher price for adults and a lower price for students and seniors. Is this price discrimination? The answer is yes! Students and seniors will have a more elastic demand because they have a lower income. Adults will have a more inelastic demand, so firms can increase their revenue by charging a lower price to students and a higher price to adults. Can you think of other examples of price discrimination?

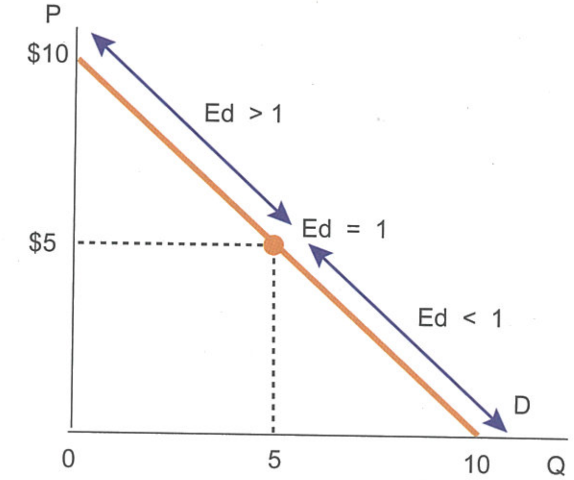
Firms can use elasticity to price discriminate between customers - increase price to customers with inelastic demand and reduce price to customers with elastic demand.

Figure 3.5 depicts a linear demand curve and shows the relationship between price elasticity and total revenue as we move along the demand curve. At the top of the demand curve where the demand curve intersects the price axis, price elasticity equals infinity. At the bottom of the demand curve where the curve intersects the quantity axis, price elasticity equals zero. At the midpoint of the demand curve, price elasticity exactly equals one. Notice that we can now divide the demand curve into the elastic segment (the top half of the D curve) and the inelastic segment (the bottom half of the D curve). If a firm is located on the top half of the demand curve it can increase its revenue by lowering price, while if it was located on the bottom half it could raise total revenue by raising price. Where is total revenue maximised on the demand curve? The answer is at the midpoint - where demand is unitary elastic. Demand is price elastic between \$10 and \$5 - as price falls, total revenue increases from \$0 to \$250. Between \$5 and \$0, demand is inelastic - as price falls, total revenue decreases from \$250 to \$0. At the midpoint (\$5), demand is unitary elastic and total revenue is maximised at \$250.

Figure 3.5 Elasticity along a linear demand curve

As you move down along a demand curve, price elasticity falls. At the top of the D curve where quantity is zero, elasticity is infinite and at the bottom of the D curve where price is zero, elasticity is zero. At the midpoint of the D curve, elasticity is equal to one.

$ Price	Quantity demanded	$TR	Ed
10	0	0	∞
9	10	90	9.00
8	20	160	4.00
7	30	210	2.33
6	40	240	1.50
5	50	250	1.00
4	60	240	0.67
3	70	210	0.43
2	80	160	0.25
1	90	90	0.11
0	100	0	0



The determinants of price elasticity of demand

The key factors determining whether a good is price elastic or inelastic are:

- **The availability of substitutes**

The greater the number of close substitutes a good has, the more price elastic its demand. If the price of good X rises and it has many close substitutes, then consumers will be sensitive to the price change because they can easily switch to other products. The demand for goods that have few substitutes such as food or water would be very inelastic. Consider the example of petrol. The demand for petrol would be highly inelastic because it has few substitutes. Once you have bought a car with a petrol engine, then you must purchase petrol to use the car - you can't fill up your tank with tap water!

- **Whether the good is a necessity or a luxury**

We would expect to find that necessity type goods such as basic items of food will be more price inelastic than luxury type goods such as jewelry, designer hand bags, and French champagne. Goods such as bread, milk and rice will be relatively inelastic in demand because these are essential food groups. Petrol and water would also be considered as necessities and would be relatively insensitive to price. Habit forming and addictive goods such as tobacco and alcohol are highly price inelastic because they are perceived as necessities by people who use them.

- **Definition of the market**

The demand for a good in a broadly defined market will be more inelastic than the demand for a good in a narrowly defined market. For example, petrol is a broadly defined market whereas a particular brand of petrol is a narrowly defined market. The demand for a specific brand such as BP, Coles Express or Caltex, would be very elastic because each brand acts as a very close substitute. If a BP service station raised its price from $1.30 to $1.40 per litre, it would lose customers to a Coles service station nearby which left its price at $1.30. We can generalise by saying that, for all goods, the price elasticity of a brand is greater than the price elasticity of the good in general.

- **The proportion of income spent**

Expensive goods are likely to be relatively price elastic because they take up a larger proportion of a consumer's income or budget. Cheaper, inexpensive items on the other hand, will be relatively price inelastic. For example, if the price of a coffee were to increase from $4 to $5 - a 25 per cent increase, it is unlikely to cause a significant decrease in the quantity demanded of coffee. However, if the price of a large screen television were to increase by 25 per cent (from $2000 to $2500) we would expect this price increase to have a greater proportional effect on quantity demanded.

- **Time**

If consumers have time to respond to a price change, then demand will be more price elastic. In the very short run, demand for most commodities will be relatively inelastic because consumers do not have the time to adjust their consumption or find substitute products. As the time period increases though, it becomes easier to change consumption patterns and so demand becomes more elastic. When petrol prices jump to very high levels, most consumers cannot change their consumption in the short run - they still have to fill their tank in order to drive to work. But if the price of petrol remains high over a longer period, then people will be able respond and adjust their consumption by using public transport or switching to electric or hybrid cars.

Price elasticity of supply

Price elasticity of supply measures the responsiveness of quantity supplied to a change in price. To calculate the formula for the price elasticity of supply, we use the same formula as for price elasticity of demand, except that we use quantity supplied instead of quantity demanded:

$$Es = \frac{\text{\% change in Qs}}{\text{\% change in price}}$$

Price elasticity of supply provides information about how quickly producers can respond to a change in price. It is measured as the percentage change in quantity supplied, divided by the percentage change in price. Similar to the formula for price elasticity of demand, price elasticity of supply can be written as:

$$Es = \frac{\Delta Qs}{Qs} \;/\; \frac{\Delta P}{P}$$

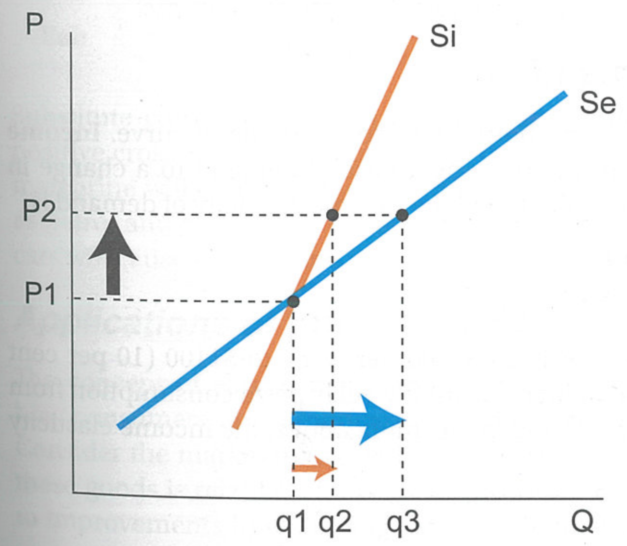
We normally transform this to:

$$Es = \frac{\Delta Qs}{Qs} \times \frac{P}{\Delta P}$$

If the percentage change in quantity supplied is greater than the percentage change in price, then supply is price elastic and the coefficient would be greater than one. For example, if the demand for pizzas increased causing the price to rise by 10 per cent and producers responded by increasing quantity supplied by 20 per cent, then the supply of pizzas would be price elastic and Es would equal 2. If on the other hand, producers could only increase quantity by 8 per cent, then supply would be price inelastic and Es would equal 0.8.

The coefficient of price elasticity of supply can also range between zero and infinity. If Es equals zero, price changes will have no effect on supply. Supply in this instance is said to be perfectly inelastic and the supply curve will be a vertical line. The supply of Rembrandt paintings, for example, is perfectly inelastic - there is a fixed quantity of originals which cannot be increased. The number of seats at the Optus Stadium in Perth is fixed at 60,000 seats, so the supply of seats is perfectly inelastic. A horizontal supply curve is said to be perfectly elastic and the coefficient Es would equal infinity. When comparing two supply curves, the steeper supply curve will always be relatively more inelastic Figure 3.6 illustrates two supply curves intersecting at the same price and quantity (P1 and Q1). When price rises from P1 to P2, quantity supplied rises for both supply curves, but by different amounts. The quantity change for supply curve Si is much smaller than the quantity change for supply curve Se. Supply curve Si is said to be more inelastic compared with Se (or supply curve Se is said to be relatively more elastic compared with Si.)

Figure 3.6 Price elasticity of supply



The supply curve Se is relatively elastic, compared to supply curve Si. Producers are more responsive to a price increase when supply is price elastic. Along Si, quantity supplied increases from q1 to q2 when price increases from P1 to P2. But along Se, quantity supplied increases by more from q1 to q3.

The determinants of price elasticity of supply

The key factors determining the responsiveness of producers are:

- **Time**

If the producer can respond quickly to a price change then supply will be price elastic. In the very short run, it may be difficult for a producer to suddenly increase output, especially if inventories are low. Supply will therefore tend to be price inelastic. As time increases, producers will be able to obtain more inputs and expand output more easily. As time increases, the supply of most goods becomes more elastic. Consider the supply of oil. At a particular point in time, the supply of oil is relatively fixed - there is a known quantity of oil reserves, so in the short term the supply of oil is relatively inelastic. However over time, exploration and discovery can increase the quantity of oil, so that in the long term the supply of oil is relatively more elastic.

- **Nature of the industry**

The supply of agricultural products tends to be relatively price inelastic, while the supply of manufactured goods is more price elastic. Products such as wheat, wool and meat require a reasonable amount of time to produce (up to 12 months). If the price of wheat suddenly increases, farmers cannot quickly respond, they must wait for the next growing season. Manufactured goods on the other hand are relatively easy to produce. Firms can quickly expand the output of computers, iPads, mobile phones and DVDs in response to an increase in price.

- **Ability to store inventories**

Inventories refer to stocks that a producer keeps stored for future sale. If a producer has the ability to store or warehouse its goods, then it can respond fairly quickly to an a change in demand and so supply would be relatively elastic. Supermarkets are able to store non-perishable goods in large warehouses and ship them whenever a store runs out of product. Goods that are perishable, such as fresh fruit and vegetables, cannot be stored readily and so their supply would be relatively inelastic.

Income elasticity of demand

There are two other types of elasticity - both linked to the demand curve. **Income elasticity of demand** (Ey) refers to the responsiveness of demand to a change in consumer income. The formula for calculating the income elasticity of demand is:

$$Ey = \frac{\text{\% change in quantity}}{\text{\% change in income}}$$

Suppose a person's income increased from \$1000 per week to \$1100 (10 per cent increase), and that person decided to increase their weekly pizza consumption from 3 to 4 pizzas (33 per cent increase). Using the formula above, the income elasticity (Ey) would be:

$$Ey = \frac{33\%}{10\%}$$

$$Ey = 3.3$$

Income elasticity can be either positive or negative. A normal good has a positive coefficient, while an inferior good has a negative coefficient.

For this consumer, pizzas are a normal good and are income elastic. Economists distinguish between normal goods and inferior goods on the basis of their income elasticities. A **normal good** has a positive income elasticity ($Ey > 0$). As income rises, demand increases. Normal goods can be further divided into those which are relatively income elastic ($Ey > 1$) and those which are relatively income inelastic ($0 < Ey < 1$). Income elastic goods are often luxury goods such as restaurant meals, overseas travel, champagne and caviar. Income inelastic goods tend to be necessities such as bread, milk, electricity and water. If you were in business would you rather sell gods that are income elastic or income inelastic?

An **inferior good** has a negative income elasticity ($Ey < 0$). As incomes increase, demand for the product actually falls. Inferior goods include plain wrap, generic brand products. Brands that are considered to be of basic, low quality would be regarded as inferior goods. Income elasticity can be useful in predicting which industry sectors will grow as incomes in the community rise. The growth in industries such as tourism, personal fitness and mobile phones reflects their strong income elasticity. Often goods that are income elastic are also price elastic (e.g. luxury goods) and goods that are income inelastic are also price inelastic (e.g. necessity type goods).

Cross elasticity of demand

Cross elasticity of demand (Eab) measures the responsiveness of the demand for one good (a) to a change in the price of a related good (b). Cross elasticity can reveal whether goods are likely to be substitutes or complements. The formula for calculating the cross elasticity of demand is:

$$Eab = \frac{\text{\% change in Qa}}{\text{\% change in Pb}}$$

Substitute goods have a positive cross elasticity while complements have a negative cross elasticity

Substitute commodities such as butter and margarine, or tea and coffee, have a positive cross elasticity of demand. If the price of butter rises, then the demand for margarine will increase (move in the same direction). Complementary goods such as petrol and cars have a negative cross elasticity of demand. A rise in the price of cars will cause the demand for petrol (its complement) to decrease.

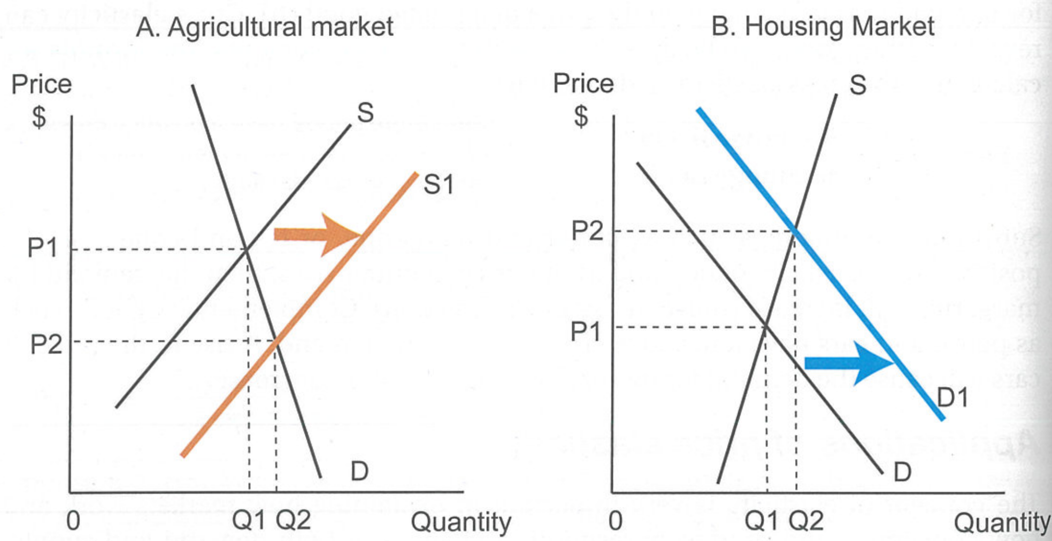
Applications of price elasticity

The concept of elasticity is very important in explaining how markets work and how consumers and producers respond to changes in both demand and supply. Consider the market for agricultural goods (Panel A in figure 3.7). The demand for these goods is relatively inelastic (steep D curve). Over time, supply increases due to improvements in technology and productivity. An increase in supply will result in price falling and quantity rising, but because demand is inelastic, total income received by the agricultural sector falls. The reason is that when demand is inelastic, an increase in production results in a large fall in price, but only a small increase in quantity sold. Farming communities have tended to contract over time because of the nature of elasticity.

Panel (b) depicts the housing industry. The supply curve for housing is very inelastic because it takes a considerable time to construct houses. Over time, demand increases due to rising incomes and rising population. An increase in demand leads to a rise in price and an increase in quantity, but when supply is inelastic, most of the effect is concentrated on price. The price of housing increases substantially, while quantity only increases modestly. The purpose of these two examples is to show that having some knowledge of price elasticity can be useful in predicting price and quantity changes over time.

Another important application of price elasticity concerns the analysis of government taxes on goods and services. Taxes on goods and services are an important source of government revenue. In Australia there is a goods and services tax (GST) levied on most goods and services and excise duties which are levied on petrol, alcohol and tobacco. When a tax is placed on a good it creates a wedge between the price paid by consumers and the price received by producers. A tax will increase the price paid by consumers, decrease the price received by producers and decrease the quantity sold. Taxes obviously have a negative impact on an industry. The GST is a very broad consumption tax covering most goods and services and is set at a level of 10 per cent.

Figure 3.7 Applications of price elasticity



When demand is inelastic, an increase in supply results in a large fall in P but only a small rise in Q.

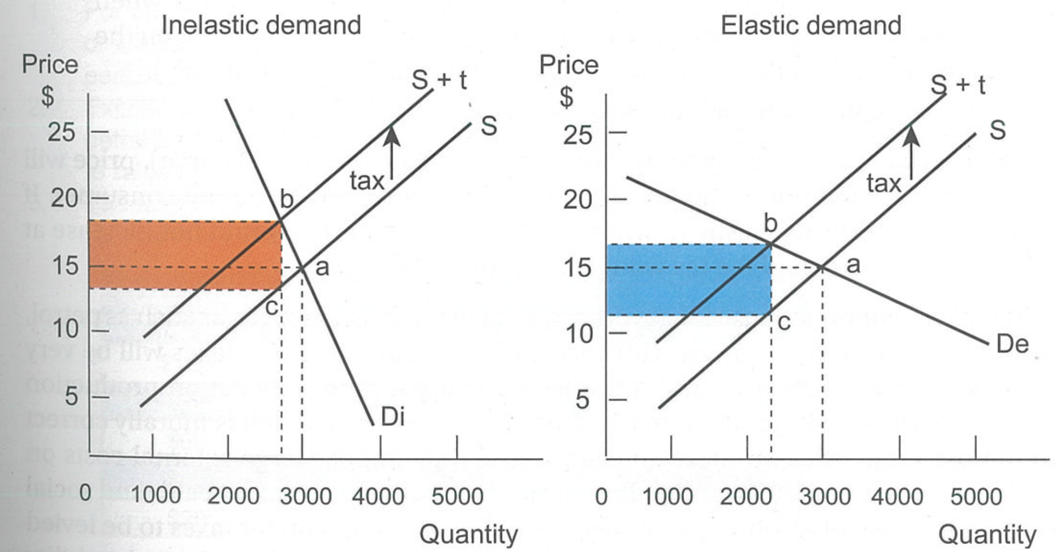
When supply is inelastic, an increase in demand results in a large rise in P but only a small rise in Q.

Excise taxes are much larger and are directed at goods with very inelastic demand, such as petrol, alcohol and tobacco. Governments often justify large taxes on these goods because they can impose significant health and environmental costs on society. Taxing goods that are inelastic in demand have a number of advantages. First, because demand is inelastic, a tax will be very effective in raising revenue. Second, taxes on inelastic goods will have a marginal impact on quantity - this means that the effect on industry output and employment will not be as great compared to a tax on a good with an elastic demand.

The effect of placing a tax on a good is shown in Figure 3.8. Panel A shows a market where demand is relatively inelastic, while panel B shows a market with a relatively elastic demand curve. A tax is a cost to the producer and is shown as a decrease in supply. The supply curve will shift by the amount of the tax. Initially the market is in equilibrium at a price of $15 and a quantity of 3000 units. Suppose that the government levies a $5 sales tax on each unit sold. The tax will cause the supply curve to shift up vertically by $5 - the supply curve shifts from S to S+t.

What is the effect of the tax? The tax has shifted the market equilibrium from point a to point b in both panels of figure 3.8. Applying a tax to a good will always increase the equilibrium price and decrease the equilibrium quantity. Will a $5 tax cause the price to increase by $5? The answer is no! Price will rise but always by less than the size of the tax. Why? Because the demand curve is downward sloping. How much the

Figure 3.8 Taxes and price elasticity



Before the tax, price is $15 and quantity exchanged is 3000. A $5 tax shifts the supply curve to S + t. After the tax, buyers pay $18 and sellers receive $13. Quantity sold falls to 2800.

Before the tax, price is $15 and quantity exchanged is 3000. A $5 tax shifts the supply curve to S + t. After the tax, buyers pay $17 and sellers receive $12. Quantity sold falls to 2300.

price rises due to the tax will depend on price elasticity. When demand is inelastic, the increase in price is much greater than when demand is elastic. How much will quantity sold fall? Again it will depend on price elasticity. The fall in quantity will be much smaller when demand is inelastic than when demand is elastic.

In the case where demand is inelastic, the after tax price has increased from $15 to $18 and the after tax quantity has fallen from 3000 to 2800. Consumers pay the new price of $18 but producers only receive $13 - the difference is the $5 tax which goes to the government. Who bears the burden or the incidence of the tax? Notice that consumers pay an extra $3 while producers receive $2 less. Most of the burden of the tax falls on the consumer because demand is inelastic. The tax revenue the government receives is shown by the shaded rectangle. It amounts to $14,000 - the tax ($5) multiplied by the quantity sold (2800).

In the case where demand is elastic, the after tax price has increased from $15 to $17 and the after tax quantity has fallen from 3000 to 2300. Consumers pay the new price of $17 but producers only receive $12 - the difference is the $5 tax which goes to the government. Who bears the burden or the incidence of the tax in this instance? Notice that consumers now only pay an extra $2 while producers receive $3 less. Most of the burden of the tax falls on the producer because demand is relatively elastic. The tax revenue the government receives is shown by the shaded rectangle. It amounts to $11,500 - the tax ($5) multiplied by the quantity sold (2300).

Notice that both the incidence of the tax and the amount of tax revenue is determined by price elasticity:

- The incidence or burden of a tax will fall more on the consumer when demand is relatively inelastic compared with supply (or fall more on the producer when demand is relatively elastic compared with supply).
- Tax revenue will be greater on goods with inelastic demand.

In the special case when demand is perfectly inelastic (vertical D curve), price will rise by the full amount of the tax and all the incidence will be on the consumer. If demand was perfectly elastic (horizontal D curve) then price would not increase at all and all the burden of the tax would fall on the producer.

So it makes economic sense for governments to impose taxes on goods such as petrol, alcohol and cigarettes - goods with very inelastic demand. These taxes will be very effective in raising revenue and they also will have a modest effect on production and employment within the industry. One could also argue that it is morally correct to impose large taxes on these goods because they impose large external costs on society - carbon emissions and pollutants in the case of petrol, and health and social costs in the case of alcohol and tobacco. Would consumers prefer taxes to be levied on elastic or inelastic goods? In the case of elastic goods, consumers have the ability to switch to a substitute good. Notice in figure 3.8 when a tax is levied on an elastic good, the price rises by only a small amount. Consumers, in other words, can easily avoid the tax by switching to a substitute good. In the case of inelastic goods, such as petrol, it is not possible to find a close substitute.

Chapter 3 Review

Worksheet 1 - price elasticity of demand

1. Define price elasticity of demand.
2. Explain the difference between elastic and inelastic demand
3. If price elasticity equals 0.4, what does this indicate?
4. What is the formula for measuring price elasticity of demand?
5. Would the demand for beer be elastic or inelastic? Explain why.
6. Would the demand for orange juice be elastic or inelastic? Explain why.
7. What is the midpoint method for measuring elasticity?
8. What does a perfectly inelastic demand curve look like? Give an example.
9. What does a perfectly elastic demand look like? Give an example.
10. How is total revenue calculated?
11. What happens to total revenue if price falls and demand is elastic?
12. What happens to total revenue if price rises and demand is inelastic?
13. If demand is elastic, should a firm raise or lower price if it wants to increase its revenue?
14. At what point on the demand curve is total revenue maximised?
15. What happens to elasticity as price increases?
16. Why do hairdressers charge males a lower price than females?
17. Explain three factors that determine price elasticity of demand.

Problem solving

1. For each of the following goods, state whether demand is likely to be elastic or inelastic and explain why - food; restaurant meals; cigarettes, crayfish; perfume; chocolate; coffee; tourist airfares
2. Explain the following statement: 'If all petrol stations increased the price of petrol by 10%, demand would be inelastic. But if BP was the only company to raise price by 10%, its demand would be elastic.'
3. The following schedules show the weekly male demand and female demand for Heather's Hair Salon:

Price of Haircut	Demand (Males)	Demand (Females)	Total Demand	Total Revenue
$100	0	50		
$80	10	70		
$60	20	95		
$40	35	115		
$20	75	140		

a. Complete the total demand and total revenue columns.
b. At what price is total revenue maximised?
c. Calculate the price elasticity of demand for both males and females. Use the midpoint formula: